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Fuel system mountable to a vehicle frame rail

Abstract

A fuel system is provided that includes a fuel system frame and in some cases access steps. The frame can be mounted to a vehicle frame rail. Bracket assemblies can be coupled to the fuel system frame at a plurality of positions. The fuel tank can be mounted at neck portions thereof and can be supported on the frame rail between the neck portions, e.g., spaced a distance from the neck portions in a longitudinal direction of the fuel system. The access steps can be non-rectangular to provide a wide stepping portion even if the fuel system includes large tanks. The steps can be directly supported by an outside surface of the tank.

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Background/Summary

CROSS-REFERENCE (1) This application is a continuation of U.S. patent application Ser. No. 17/149,466, filed Jan. 14, 2021, which is a continuation of PCT Application No. PCT/US2020/023154, filed Mar. 17, 2020, which claims the benefit of U.S. Provisional Application No. 62/822,439, filed Mar. 22, 2019. Each of the foregoing applications is hereby incorporated by reference in its entirety herein.

BACKGROUND OF THE INVENTION

Field of the Invention

(1) This application relates to fuel systems that can be mounted to a lateral or side portion of a vehicle frame rail.

Description of the Related Art

(2) Compressed natural gas (CNG) is an alternative fuel that provides many advantages. CNG fuels burn cleaner than other combustion fuels for vehicles. CNG also can be more cost effective.

(3) CNG fuel systems can come in several forms. One form employs a Type IV fuel tank constructed with a polymeric liner. Carbon fiber wrapped around the liner can reinforce the liner, to produce a fuel tank strong enough for use on heavy-duty trucks and other vehicles. The fuel tank can have a boss sealing each of the end portion of the fuel tank. The boss can provide access to the fuel tank for filling and dispensing the fuel contained therein. A side mounted fuel system can include a frame to support the fuel tank on a side or lateral portion of a vehicle. Straps can support central portions of the fuel tank within the frame. Some fuel tanks also can be supported at one or both ends at the bosses.

SUMMARY OF THE INVENTION

(4) While the side mounted fuel system is known, complications in mounting the fuel system can arise. Straps adds cost, complexity, and a failure mode to the fuel system. Also, other components are mounted to the lateral portion of the frame rail. So it can be challenging to locate the fuel system conveniently relative to these other components. A need exists to provide improved side-mounted fuel systems. There is a need for improved assemblies and systems that can be more flexibly connected to the frame rail, e.g., a two or more positions spaced along the frame rail. There is a need for improved assemblies and systems that support a fuel tank at a boss portion. Also, there is a need for an improved cab access system. These improvements can enable larger tanks to be supported to a lateral portion of a frame rail while not extending the width of the vehicle at the tank beyond acceptable limits.

(5) In one embodiment, a fuel system is provided that includes a fuel system frame, a first bracket, and a second bracket. The fuel system frame has a first position and a second position spaced apart from the first position. The first bracket is configured to connect to the fuel system frame at the first position or at the second position to support the fuel system from either the first position or the second position. The second bracket has a first portion configured to mate with the first bracket and a second portion configured to connect to a vehicle frame rail.

(6) The first position and the second position allow the first and second brackets to be located at any one of a plurality of different vehicle frame rail positions along the frame rail without requiring the movement of the fuel system frame.

(7) The first position and the second position allow the fuel system to be located in any one of a plurality of different vehicle frame rail positions along the frame rail without requiring the movement of the second bracket.

(8) In another embodiment, a fuel system is provided that includes a fuel tank and a support assembly. The fuel tank includes a central cylindrical portion, a first neck portion that has a first boss at a first end and a second neck portion that has a second boss at a second end. The support assembly is configured to connect the fuel system to a side portion of a frame rail of a vehicle. The support assembly has a first tank support portion, a second tank support portion, and a bracket system. The first tank support portion is configured to support the first neck portion of the tank. The second tank support portion is configured to support the second neck portion of the tank. The bracket system is coupled with the first tank support portion and the second tank support portion. The bracket system is configured to be coupled to a frame rail at a frame rail position longitudinally between the first neck portion of the fuel tank and the second neck portion of the fuel tank.

(9) In some variations, the tank is not being directly supported at locations longitudinally between the first and second neck portions.

(10) In another embodiment a fuel system is provided that includes a fuel tank, a frame, and a step support. The fuel tank has a central cylindrical portion, a first end and a second end opposite the first end. The frame has a frame rail connection portion disposed on a vehicle side of the fuel system and a tank support portion disposed around one or both of the first and second ends of the fuel tank. The step support is configured to apply a load to the central cylindrical portion of the fuel tank.

(11) In some embodiments, a fuel system is provided that includes an access door on an end of the fuel system. The access door can be configured to be moved to provide access to three or more components within the fuel system. One of the three or more components can be a filter. For example, in some embodiments, the access door can provide access to a filter, a defuel assembly, and a bleed valve. In some embodiments, the access door can provide access to a filter, a cylinder valve assembly, and a bleed valve.

(12) In some configurations the fuel system enclosure, e.g., cover, has an elongated side opening that provides access to one or more components. The components can be mounted to one or more modular frame members, e.g., plates. The elongate opening enables service access to one or more components that may be mounted in a modular way and accessible through the opening. For example, the same exterior enclosure with the elongate access opening can be used to enclose a wide range of internal component layouts while providing service, inspection and repair access to the components thereof. In one example, three modular frame members are provided, each with one or more serviceable component mounted thereto. The access opening can provide access to the components on all three modular frame members.

(13) In another form, an enclosure is provided with an access door assembly having two stop positions. The assembly can have a first stop position corresponding to the door being rotated out of a position of covering an access opening. The assembly can have a second stop position corresponding to the door being rotated to a position of covering the access opening. In one of the positions, a hook comes to rest on a shaft of the access door assembly whereby the position of the door is restrained. The door can be fixed in the restrained position by actuating a door locking clamp.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The abovementioned and other features of the inventions disclosed herein are described below with reference to the drawings of the preferred embodiments. The illustrated embodiments are intended to illustrate, but not to limit the inventions. The drawings contain the following figures.

(2) FIG. 1 is a perspective view of a cab of a heavy duty vehicle having a fuel system mounted to a side portion of a frame rail of the vehicle;

(3) FIG. 2 is a top view of a portion of a vehicle frame rail of a vehicle chassis having a fuel system according to one embodiment coupled to a lateral portion of the frame rail;

(4) FIG. 3 is a disassembled view of a portion of a fuel system bracket and a portion of a fuel system support assembly that has a plurality of connection positions;

(5) FIGS. 3A-3B show a tank-side and frame rails side views, respectively, of a bracket assembly including a bracket portion and a fuel system support assembly portion having a continuous range of connection positions;

(6) FIG. 4 is a perspective view of a frame rail and bracket assembly for mounting a fuel system according to one embodiment;

(7) FIG. 5 is a vehicle side view of a fuel system and bracket assembly according to one embodiment;

(8) FIG. 6 shows detail 6-6 of the fuel system and bracket assembly of FIG. 5;

(9) FIG. 7-8 are top and perspective views of a bracket assembly according to one embodiment;

(10) FIG. 9 is a cross-section of the bracket assembly of FIG. 2 taken at the section plane 9-9 showing an elevation support of a fuel system bracket;

(11) FIG. 10 is a cross-section of the bracket assembly of FIG. 2 taken at the section plane 10-10 showing a transverse motion limit portion of a frame rail bracket;

(12) FIG. 11 is a perspective view of the fuel system of FIG. 2 with the cover thereof removed

showing neck support of the fuel tank;

(13) FIGS. **11A-11B** show a perspective view of an alternative configuration in which fuel system mounting positions are located outside of a projection of the fuel tank, e.g., of a forward end of the fuel tank and/or rearward of a rearward end of the fuel tank;

(14) FIG. **12** is a perspective view of a frame assembly of the fuel system of FIG. **2** with the cover and the fuel tank removed;

(15) FIG. **13** is an end view of the fuel system of FIG. **2** with an end portion of the cover of the system removed, showing a step assembly of the fuel system;

(16) FIG. **13A** is a detail view of a step assembly of FIG. **13** of the fuel system of FIG. **2**;

(17) FIG. **14** is an end view of the fuel system of FIG. **2** showing an access door in an open position;

(18) FIG. **14A** is a perspective view of an end of the fuel system of FIG. **2** showing the access door of FIG. **14** in an open position; and

(19) FIG. **14B** is an end view of the fuel system of FIG. **2** showing the access door of FIG. **14** in a closed position.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENT

(20) While the present description sets forth specific details of various embodiments, it will be appreciated that the description is illustrative only and should not be construed in any way as limiting. Furthermore, various applications of such embodiments and modifications thereto, which may occur to those who are skilled in the art, are also encompassed by the general concepts described herein. Each and every feature described herein, and each and every combination of two or more of such features, is included within the scope of the present invention provided that the features included in such a combination are not mutually inconsistent.

(21) This application discloses a number of improvements in a side mounted fuel system **100** that, in some embodiments, provides more options in mating or mounting the fuel system **100** to a frame rail **42** of a vehicle **40**. See, e.g., FIG. **2** and Section I below. In some cases, the improvement provide enhanced neck mounting of a fuel tank **102** of the fuel system **100**. See, e.g., FIGS. **11-12** and Section II below. In some cases, the improvements make better use of available space for vehicle access steps a large fuel tank **102**. See, e.g., FIG. **13** and Section III below.

(22) I. Support Assembly for Frame Rail Connections

(23) Some embodiments of the fuel system **100** provide more options for how mounting the fuel system **100** to a vehicle **40**. The vehicle **40** can be or include a portion of a heavy-duty vehicle, such as a tractor unit for forming a tractor-trailer. The vehicle **40** includes a frame rail **42** that supports various components, such as forward wheels, rearward wheels, and a cab. The frame rail **42** can include a rigid member formed into a shape providing enhanced strength, such as having an I-beam or a C shaped cross-section configuration. A C-shaped frame rail **42** can have a long side **44** facing away from a center of the frame rail **42** and short sides at a top and a bottom side thereof. The side **44** faces away from a central, vertical forward-rearward plane. The side **44** can be equipped to support the fuel system **100** as well as a number of other components.

(24) FIG. **2** shows an example of how the fuel system **100** and other components can be mounted to the frame rail **42**. The fuel system **100** accommodates these other components to make better use of restricted space and/or to provide compact configurations. For example, a fender **46** can be disposed at least partially over front wheels of the vehicle **40**. In one embodiment, a gap **47** separates a forward face of the fuel system **100** from a rearward edge of the fender **46**. A bracket assembly described below can adjust the gap **47**. As a further example, a shackle **50** coupled to the frame rail **42** can be configured to support a shock absorber. The shock absorber configuration and/or the geometry of the vehicle **40** can limit a desired position for the shackle **50**.

(25) The mounting location of the fuel system **100** can be shifted to better accommodate these other vehicle components. Also, the bracket assembly component of the fuel system **100** to be shifted even if the rest of the fuel system **100** is not shifted to accommodate these other components.

(26) A. Frame Rail Mounting Location Adjustment

(27) FIGS. 2, 3, 5, and 11 show further aspects of mounting location for the fuel system 100. The fuel system 100 includes the fuel tank 102 that is supported by a support assembly, e.g. a fuel system frame 104. The fuel system frame 104 can also include a frame rail connection portion 106 that can couple with a portion of a bracket assembly in a number of different positions. The frame rail connection portion 106 is provided on a vehicle side 100V of the fuel system 100. The frame rail connection portion 106 can have a grid of mount features, which can include a plurality of mounting holes as discussed further below. The plurality of mounting holes enables greater flexibility in placement of the brackets used to connect the fuel system 100 to the frame rail 42. The plurality of mounting holes enables greater flexibility in placement of the fuel system 100 on the frame rail 42. The plurality of mounting holes enables greater flexibility in placement of the fuel system 100 or the brackets along the frame rail 42. The fuel system 100 also includes a cover 105 that in combination with the fuel system frame 104 encloses a space around the fuel tank 102.

(28) FIGS. 2, 3 and 6-8 show details of a first bracket 108 and the connection thereof to the frame rail connection portion 106 at a rearward portion of the fuel system 100. In one embodiment, the frame rail connection portion 106 is able to couple with a first bracket 108 in more than one position. The first bracket 108 can be coupled to a second bracket 112 to mount the fuel system 100 to the frame rail 42 as discussed further below. FIG. 3 shows that in one example, the frame rail connection portion 106 can have an array of holes that allows the first bracket 108 to couple to the frame rail connection portion 106 in a plurality of discrete predefined positions. A first frame rail connection portion 106 can be coupled to a rearward position of the fuel system 100 in the orientation shown in FIG. 3 and a second frame rail connection portion 106 can be coupled to a forward position of the fuel system 100. The second frame rail connection portion 106 toward the forward position can be in an inverted orientation (rotated 180 degrees about an axis into and out of the page). As a result, mounting features 107 (e.g., fastener holes) of the rearward frame rail connection portion 106 can be located toward the rear of the fuel system 100 and the mounting features 107 of the forward frame rail connection portion 106 can be located toward the front of the fuel system 100.

(29) A first position 120 can be provided by a subset of the holes on the frame rail connection portion 106. For example a pair of holes located toward a forward end of the array of holes, one above the other, can in part define the first position 120. The forward pair of holes can be located at a common longitudinal position of the frame rail connection portion 106. A second pair of holes can in part define the first position 120. The second pair of holes of the first position 120 can be located at a common longitudinal position of the frame rail connection portion 106 spaced apart from the first pair of holes of the first position 120. The first position 120 includes in this embodiment as a set of four holes including a forward most pair of holes on the frame rail connection portion 106. This configuration allows the frame rail connection portion 106 to be coupled with the first bracket 108 at a forward-most position of the frame rail connection portion 106. Although the first position 120 can include four holes, in some embodiments the first position 120 can include more or fewer holes than four holes, can include a different arrangements of holes, and can include positions along one or more slots as described further below. This can allow the first bracket 108 and a second bracket 112 coupled therewith to be located forward relative to a second position 124 defined by a set of holes of the array of holes on the frame rail connection portion 106.

(30) In one embodiment, the second position 124 is defined by a pair of holes disposed toward the rearward end of the frame rail connection portion 106. The pair of holes of the second position 124 disposed toward the rearward end can be disposed at a common longitudinal position of the frame rail connection portion 106. The second position 124 can be further defined by a second pair of holes spaced forward of the first pair of holes of the second position 124. The second pair of holes of the second position 124 can be located forward of, and in the illustrated example immediately adjacent to, the rearward pair of holes of the first position 120. Although the second position 124

include four holes, the second position **124** could include more or fewer holes than four holes, can include different arrangements of holes, and can include positions along one or more slots as described further below.

(31) The frame rail connection portion **106** can define a third position **125** between the first position **120** and the second position **124**. The third position **125** can be defined in part by a pair of holes located rearward of, e.g., in one case immediately adjacent to, the forward holes of the first position **120**. The third position **125** can be further defined by a second pair of holes located rearward of, e.g., immediately rearward of, the rearward holes of the first position **120**. Although the third position **125** include four holes, the third position **125** could include more or fewer holes than four holes, could include different arrangements of holes, and could include positions along one or more slots as described further below. Although three positions are illustrated in the frame rail connection portion **106**, in other embodiments there can be more or fewer positions defined by discrete, predefined holes in the frame rail connection portion **106**. The positions **120**, **124**, and **125** can also be forward, rearward and intermediate in a second frame rail connection portion **106** which can be inverted and can be located toward the forward end of the fuel system **100** as discussed above.

(32) The distance between the positions **120**, **124**, **125** can be any suitable distance. For distance from the forward pair of holes of the first position, **120** to the forward pair of holes of the second position **124** can be 100 mm in one embodiment. The distance from the second position **124** to the third position **125**, e.g., from the forward pair of holes of the first position **124** to the forward pair of holes of the second position **125** can be 50 mm in one embodiment. These distances can be uniform or can vary from one position to the next. For example, the distance from the forward pair of holes of the first position **120** to the forward pair of holes of the second position **124** can be any of 200 mm, 150 mm, 100 mm, 50 mm, 25 mm or more or less. The distance from the second position **124** to the third position **125**, e.g., from the forward pair of holes of the first position **124** to the forward pair of holes of the second position **125** can be any of approximately 160 mm, 155 mm, 152 mm, 143 mm, 126 mm, 100 mm, 75 mm, 70 mm, 61 mm, 50 mm, 32 mm, 25 mm, or 12 mm. The distance from the second position **124** to the third position **125** can be three-quarters, two-thirds, one-half, one-third or one-quarter of the distance from the first position **120** to the second position **124**.

(33) FIG. 3 can further illustrate the positioning of a bracket assembly relative to the frame rail connection portion **106**. A first bracket **108** can include a portion illustrated in FIG. 3 that is configured to be coupled with the frame rail connection portion **106**. The portion of the first bracket **108** can include a central flange **164**. The central flange **164** can project away from a bracket mounting base **166** that can include a plurality of mounting holes **109**. The mounting holes **109** can be arranged to match the holes corresponding to one or more position of the frame rail connection portion **106**. For example, the mounting holes **109** can be arranged in an array that includes a pair of mounting holes **109** on a first side of the central flange **164** and a pair of mounting holes **109** on a second side of the central flange **164** opposite the first side. Thus, the mounting holes **109** can be aligned over the holes of the first position **120** and fasteners can be used to secure the bracket mounting base **166** to the frame rail connection portion **106**.

(34) Advantageously, the groups of holes forming the first position **120**, the second position **124**, and the third position **125** can all be arranged at the same spacing as the mounting holes **109** such that the bracket mounting base **166** can be aligned with the holes of any of these positions.

(35) As noted above, the fuel system frame **104** can include the frame rail connection portion **106** located toward a rearward portion of the fuel system **100**. The fuel system frame **104** can also include a second frame rail connection portion **106** located toward a forward portion thereof. A structural member **256** can couple the two frame rail connection portion **106** together or they can be formed on a single unitary frame member. The forward frame rail connection portion **106** can have the same or a similar configuration as the rearward frame rail connection portion **106**, e.g., with an

array of holes defining a plurality of discrete spaced apart positions. The forward frame rail connection portion **106** can be coupled with bracket mounting base **166** of a third bracket **140**, the third bracket **140** configured to couple with a fourth bracket **152** to form a second bracket assembly to couple the fuel system **100** to the frame rail **42**.

(36) The holes in the frame rail connection portions **106** (forward and rearward) enable several types of mounting configurations. For example, in one case, the longitudinal position of the fuel system **100** along the frame rail **42** can be determined. For example, the desired gap **47** can be provided between a forward portion of the fuel system **100** and an adjacent component, such as the fender **46**. In another example, the vehicle side **100V** of the fuel system **100** can be disposed at least partially over a low profile component, such as the shackle **50**. Potential interference at the forward end of the fuel system **100** with other components on the frame rail **42** can be determined. In one example, securing the third bracket **140** to the frame rail connection portion **106** in the first position **120** would result in interference with another such component. The third bracket **140** can be moved to the second position **124** or the third position **125** of the forward frame rail connection portion **106**. This allows the fourth bracket **152** to move to a frame rail position corresponding to the second position **124** or the third position **125** out of interfering position with such other component.

(37) FIG. 2 shows the third bracket **140** in the second position **124** of the forward frame rail connection portion **106**. FIG. 3 shows that in the case of the rearward frame rail connection portion **106** the first position **120** is a forward position, the second position **124** is a rearward position, and the third position **125** is an intermediate position between the forward and rearward positions. In the case of the forward frame rail connection portion **106**, the second position **124** is also the rearmost position of the forward frame rail connection portion **106**. For the forward frame rail connection portion **106** the second position **124** is the position farthest from the mounting features **107**. The position of the first bracket **108** on the rearward frame rail connection portion **106** can also be selected to avoid interference with other components on the frame rail **42**. In the illustrated embodiment, the bracket mounting base **166** of the first bracket **108** can be secured to the fuel system **100** at the first position **120**, forward of the other positions **124**, **125**. This enables the rearward bracket assembly including the first bracket **108** and the second bracket **112** to move forward out of a potentially interfering position with other component on the frame rail **42**. One can see from the foregoing description that a number of permutation is possible. In the illustrated embodiment, there are nine permutations that are possible for positioning the first bracket **108** and the third bracket **140**. These positions can result in corresponding shifting of the second bracket **112** and the fourth bracket **152** as needed to provide convenient connection, e.g., out of interference with other components connected to the frame rail **42**.

(38) FIGS. 3A and 3B illustrate another embodiment of a frame rail connection portion **106A** that can be integrated into the fuel system frame **104**. The frame rail connection portion **106A** can facilitate adjustment along a continuous range of positions that are not predefined within the range. The frame rail connection portion **106A** enables connection of the bracket mounting base **166** of the first bracket **108** or of the third bracket **140** to the rearward frame rail connection portion **106A** and to the frame rail connection portion **106A**. In one embodiment, the frame rail connection portion **106A** includes a first slot **126** and a second slot **127**. The first slot **126** can be located generally above the second slot **127**, e.g., extending parallel to the second slot **127**. Fasteners disposed through the mounting holes **109** of the bracket mounting base **166** of the first bracket **108** can be located at any position along the first slot **126** and second slot **127**.

(39) The position of the first bracket **108** can be defined by initially loosely coupling the first bracket **108** to the frame rail connection portion **106A** at the first slot **126** and/or the second slot **127**, e.g., by bolts or other fasteners. The position of the first bracket **108** can be refined by sliding the first bracket **108** along the frame rail connection portion **106A** to reach a user-defined position that is beneficial, e.g., that avoids interference as needed. This allows positioning of a bracket

assembly including the first bracket **108** to be selected during mounting to the vehicle **40** at any such position. In a fuel system frame **104** that includes forward and rearward frame rail connection portion **106A**, the third bracket **140** also can be coupled and positioned in this manner, allowing the end user to select the positions along the first slot **126** and second slot **127** during mounting of the fuel system **100** to the vehicle **40**.

(40) B. Bracket Assemblies Providing Fastenerless Secure Connections

(41) FIGS. **4-10** illustrate bracket assemblies that provide convenient connection during fuel system-to-vehicle assembly. These bracket assemblies provide secure connections without requiring the bracket interface to be further secured by bolts or other similar fasteners. Such fasteners can be provided prior to final shipment.

(42) FIG. **4** shows that in one assembly a plurality of, e.g., two vehicle-side bracket assembly portions can be coupled to the frame rail **42**. The second bracket **112** can be secured to the side **44** of the frame rail **42** forward of the fourth bracket **152**. The second bracket **112** can form a portion of a rearward bracket assembly with the first bracket **108**. The fourth bracket **152** can form a portion of a forward bracket assembly with the third bracket **140**. The brackets **112**, **152** have similar construction so the description of each one applies to the other. The second bracket **112** includes a first portion **128** and a second portion **132**. The second portion **132** facilitates connection to the side **44** of the frame rail **42**. For example, the second portion **132** can have an array of mounting holes that can receive fasteners to be secured across the frame rail **42**. The first portion **128** can be a continuous member with the second portion **132** or can be part of an assembly therewith. The first portion **128** can allow the first bracket **108** to be lowered onto the second bracket **112**. The first portion **128** can support the first bracket **108** so that the fuel system **100** coupled therewith is retained on the frame rail **42**. A flange member can be provided on the first bracket **108** can come to rest on top of the second portion **132** of the second bracket **112**. A portion of the first bracket **108** can come to rest against a lower surface of the second portion **132** of the second bracket **112**. The fourth bracket **152** can have a first portion **156** similar to the first portion **128** and a second portion **160** similar to the second portion **132**.

(43) FIGS. **5-6** show details of the first bracket **108**. The first bracket **108** can be located rearward of the third bracket **140** on the vehicle side **100V** of the fuel system **100**. The position of each of the brackets **108**, **140** can be selected by the user, e.g., based on the desire to avoid interference with other components, as discussed above. In the illustrated embodiment, the first bracket **108** is in the first position **120** of the rearward frame rail connection portion **106** and the third bracket **140** is in the second position **124** of the forward frame rail connection portion **106**. The third bracket **140** can have the same construction as the first bracket **108**.

(44) FIG. **6** shows the structure of the first bracket **108** in more detail. The first bracket **108** includes a body including the central flange **164** and the bracket mounting base **166**. The bracket mounting base **166** can include a generally planar side configured to be secured against the frame rail connection portion **106**. The central flange **164** can extend away from a side of the bracket mounting base **166** opposite the side configured to mate with the frame rail connection portion **106**. The central flange **164** can have opposite sides that face forward and rearward when the first bracket **108** is mounted to the frame rail connection portion **106** and to the frame rail **42**. The first bracket **108** can include a mounting portion for coupling the first bracket **108** to the second bracket **112**. The mounting portion can include a bracket aperture **167** disposed along an axis extending between the opposing sides of the central flange **164**. The bracket aperture **167** provides a point of connection between the first bracket **108** and the second bracket **112** as discussed further below.

(45) The first bracket **108** includes first and second support wings **168**, **172**. The support wings can have similar, e.g., mirror image, configurations. The first support wing **168** can include a mount portion **173** and a vertical support **174**. The mount portion **173** and vertical support **174** form a flange configuration for resting on top of the second portion **132** of the second bracket **112**. The mount portion **173** is configured to extend along a forward side of the central flange **164**.

(46) The mount portion **173** can be secured adjacent to the central flange **164** through one or more resilient members, e.g., vibration dampers. The vertical support **174** can extend rearward from an upper portion, e.g., the top of, the mount portion **173**. The vertical support **174** can be configured to rest on top of a portion of the second bracket **112** in a predefined position when the first bracket **108** and the second bracket **112** are connected. The position can be defined at least in part by a transverse motion limit structure. In one embodiment, at least the vertical support **174** includes a configuration to resist bending. For example, the cross-section of the vertical support **174** can have a one or more flanges that extend away from the loading surface, where forces are applied between the vertical support **174** and the second bracket **112**. The flanges act in a manner similar to an I-beam to enhance the stiffness of the first support wing **168**. The flanges can be seen as areas of greater height at the vehicle facing side and at the fuel system facing side of the first support wing **168**.

(47) The first support wing **168** also can include a rotational support **176**. The rotational support **176** can include a projection configured to rest against a lower portion, e.g., surface, of the second bracket **112**. The rotational support **176** can help to enable the fuel system **100** to be supported on the frame rail **42** with the force of gravity being countered at least in part through the rotational support **176**. The weight is further supported through the first support wing **168** and the second support wing **172**. A clearance is provided between a top surface of the rotational support **176** and a bottom surface of the vertical support **174** to receive the first portion **128** of the second bracket **112**. This is described in greater detail below in connection with FIGS. 7-10.

(48) The connection between the first support wing **168** and the second support wing **172** and the central flange **164** can be through a resilient member, such as a damper or a vibration isolator. FIG. 6 shows that a first resilient member **180** provided between the first support wing **168** and the central flange **164**. The first resilient member **180** can be configured as a stepped cylinder having a smaller diameter section on a central portion of the outside surface there. The first resilient member **180** can have larger diameter section on ends thereof on both sides of the central portion. The larger diameter section of one end of the first resilient member **180** can be disposed between opposing surfaces of the first support wing **168** and the central flange **164**. The larger diameter section of another end of the first resilient member **180** can be disposed between opposing surfaces of the second support wing **172** and the central flange **164**. As shown, bolts can be disposed through the first resilient member **180** and the second resilient member **182** to secure the first support wing **168** to the central flange **164**. The first resilient member **180** and the second resilient member **182** are resilient, e.g., compressible, such that loads applied during operation of the vehicle **40** from the road are not directly transferred to the fuel system **100** but rather are absorbed to some extent and modulated in the first resilient member **180** and the second resilient member **182**. In one variation, the first resilient member **180** and the second resilient member **182** can be combined providing a single member with multiple aperture for securing the wings to the central flange **164**.

(49) A third resilient member **184** provided in the bracket aperture **167** modulates similar loads that could be applied through the point of connection at the bracket aperture **167** between the first bracket **108** and the second bracket **112**. In one modified embodiment, connection at the bracket aperture **167** can be provided without any vibration isolation or damping, e.g. by directly bolting the first bracket **108** to the second bracket **112** at this location. In a further modified embodiment, the first resilient member **180** and the second resilient member **182** can be eliminated. The connection points between the first and/or second support wing **168**, **172** can be rigid e.g., direct connection without intervening resilient members.

(50) FIGS. 4, 9 and 10 illustrate aspects of the second bracket **112**. In one embodiment, the second bracket **112** and the fourth bracket **152** have the same construction. So, description of either the second bracket **112** or of the fourth bracket **152** applies to the other these two brackets. The second bracket **112** includes a first portion **128** and a second portion **132**, as discussed above. The first portion **128** can include a generally planar structure at least on a side configured to face the side **44**

of the frame rail **42**. The first portion **128** can have one or a plurality of apertures for securing the second bracket **112** to the frame rail **42** using fasteners. The second bracket **112** can have a vertical support **186** on each of a forward and a rearward side of an internal space **188** disposed on a lateral side of the second bracket **112**. The internal space **188** can be disposed on a side opposite the side of the second bracket **112** that faces the side **44** of the frame rail **42** when the second bracket **112** is coupled thereto.

(51) The vertical supports **186** can be formed flanges of the second bracket **112** located forward and rearward of the internal space **188**. The internal space **188** can be configured, e.g., sized and shaped, to receive the central flange **164** and the mount portion **173** of the first bracket **108**. The vertical supports **186** can be configured to reside immediately below and in contact with the first support wing **168** and the second support wing **172** when the first bracket **108** is mated with the second bracket **112**. The vertical support **186** can have a geometry to enhance stiffness, e.g., with a support flange disposed beneath the vertical support **186** in contact with the second portion **132**.

(52) The second bracket **112** can include a transverse motion limiter **196** configured to control the position of the first bracket **108** relative to the second bracket **112** when these brackets are assembled together. The transverse motion limiter **196** can include one or more, e.g., two, ridges **198** disposed along a top surface of the vertical support **186**. The ridges **198** can have a height of about 0.1 inch. In some embodiments, the ridges **198** have a height of 0.2 inch, 0.3 inch, 0.4 inch, 0.5 inch, 0.6 inch, 0.7 inch, or more than 0.7 inches. The ridges **198** can have a height that is 5% of the thickness of the first support wing **168** or the second support wing **172** in a vertical direction, e.g., of the transverse stiffening structure of the support wing. The ridges **198** can have a height that is 10% of the thickness of the first support wing **168** or the second support wing **172** in a vertical direction. The ridges **198** can have a height that is 25% of the thickness of the first support wing **168** or the second support wing **172** in a vertical direction.

(53) The support wings **168**, **172** and the vertical supports **186** can allow the fuel system **100** to be mounted to the frame rail **42** during assembly without any additional fasteners. The support wings **168**, **172** and the vertical supports **186** can have apertures for allowing bolts to more permanently connect the second bracket **112** to the first bracket **108**, as discussed further below. The second bracket **112** also can include a bracket aperture **190** configured to be aligned with the bracket aperture **167** when the brackets are assembled. The alignment of the bracket aperture **190** on the second bracket **112** and the bracket aperture **167** on the first bracket **108** can be facilitated by a clearance gap **192** formed through the second bracket **112**. The clearance gap **192** can allow a portion of the body of the first bracket **108** disposed about the bracket aperture **167** to overlap with the first portion **128** of the second bracket **112**.

(54) FIGS. 7-10 illustrate methods of assembling the fuel system **100** to the frame rail **42**. FIG. 7 shows that the fuel system **100** with the first bracket **108** attached thereto at the frame rail connection portion **106** can be placed adjacent to the second bracket **112**, which is coupled to the side **44** of the frame rail **42**. The first bracket **108** can be placed at or slightly above the elevation of the second bracket **112**. FIG. 8 shows that the first bracket **108** can be moved into the internal space **188**. For example, the portion of the central flange **164** away from the vehicle side **100V** can be moved into the internal space **188**. The mount portion **173** of the first support wing **168** can be moved into the internal space **188**. The first bracket **108** and the fuel system **100** coupled thereto can be moved down relative to the second bracket **112** from the position shown in FIG. 8 until a bottom side of the first support wing **168** (and of the wing **172**) is disposed above, on top of or in contact with the vertical support **186**. Specifically the vertical support **174** of the first support wing **168** of the first bracket **108** can be placed on top of the vertical support **186** of the second bracket **112**. A vertical support of the wing **172** of the first bracket **108** can be placed on top of the vertical support **186** of the second bracket **112**. FIG. 9 shows that when so placed, the rotational support **176** of the first bracket **108** can come into contact with a lower surface of the second bracket **112**. The lower surface can be disposed on a transverse portion of the second bracket **112** disposed

between the vertical support **186** and facing or partly bounding the internal space **188**. The rotational support **176** can limit rotation of the fuel system **100** relative to the frame rail **42**, which rotation can be a function of the outboard weight of the fuel system **100**.

(55) FIG. **10** shows that the transverse position of the fuel system **100** relative to the frame rail **42** can be at least partially controlled by the transverse motion limiter **196**, e.g., by ridges **198** located on top of the vertical support **186**. The ridges **198** can be configured, e.g., sized and positioned to abut an in-board face of the first bracket **108**, e.g., of an inboard portion of one or both of the support wings **168**, **172**. Said another way, as the fuel system **100** and the first bracket **108** move toward the frame rail **42** the in-board side of one or both of the wings **168**, **172** will abut the ridges **198** and such abutment can be detected by the assemble device or personnel positioning the fuel system **100**. This will confirm that the first bracket **108** and the second bracket **112** are at least temporarily connected. This position will also bring the bracket aperture **167** in line with the bracket aperture **190**, as discussed above.

(56) These configurations allow the fuel system **100** to be at least temporarily secured to the frame rail **42** without any additional connecting devices, e.g., without any bolts passing through both of the brackets **108**, **112**. FIGS. **9** and **10** shows that subsequently, e.g., before sending the vehicle **40** out on the road at the end of assembly, bolts can be passed through both brackets **108**, **112** to provide a more secure connection.

(57) The manner of assembling the third bracket **140** to the fourth bracket **152** can be the same as that discussed above in connection with the first bracket **108** and second bracket **112**. Additionally, the bracket assembly methods can include selecting a position from a discrete number of positions or along a continuous range, for connection of the first bracket **108** to the frame rail connection portion **106**. The position selected can be based on the presence of other components on the frame rail **42**, e.g., of a frame cross-member. The position of the first bracket **108** can follow a preferred location of the second bracket **112** to avoid interference with such components. The position of the first bracket **108** can follow a preferred location of the second bracket **112** to share a mount location with another member. The position of the second bracket **112** can follow a preferred location of the first bracket **108**.

(58) II. Neck Mounted Fuel Tank Assemblies for Side Mounted Fuel Systems

(59) The fuel system **100** provides a number of advantages, some of which relate to the manner in which the fuel tank **102** thereof is supported within the cover **105**. The fuel tank **102** is supported in novel ways, e.g. at ends thereof and with arcuate supports that provide advantages positioning relative to the frame rail **42**. In some embodiments, the fuel system frame **104** is configured with low profile mounting configurations, to maintain mounting locations between ends of the cover **105** of the fuel system **100** to provide low profile mounting configurations. In some embodiments, the fuel system frame **104** has extended mounting location configurations to position the mounting locations outside the area opposite the fuel tank **102** to leave the frame rail **42** free from connections opposite the fuel tank **102**.

(60) A. Low Profile Mounting Configurations

(61) FIG. **11** shows the fuel system **100** with the cover **105** removed. The fuel tank **102** has a central cylindrical portion **204**, a first neck portion **208**, and a second neck portion **212**. The first neck portion **208** includes a first boss **220** located at the end of the fuel tank **102**. The first boss **220** is a sealed portion of the fuel tank **102** that can have fuel conduit coupled therewith to enable filling fuel into and drawing fuel from the fuel tank **102**. The first boss **220** is also sufficiently rigid to enable the fuel tank **102** to be supported at that location as discussed further below. The fuel tank **102** can have a second boss **228** disposed at a second end **232** of the fuel tank **102**. The fuel tank **102** also can include a polymeric liner at least in the central cylindrical portion **204** thereof. The liner can be secured to a first dome assembly including the first boss **220** and to a second dome assembly including the second boss **228**.

(62) In some embodiments the fuel tank **102** is supported only at the first neck portion **208** and the

second neck portion **212**. In such embodiments, no straps are provided in the central cylindrical portion **204** nor is the fuel tank **102** supported in any other manner between the first end **224** and the second end **232**. In other embodiments, one or more supporting straps also can be provided between the first end **224** and the second end **232** to support the fuel tank **102**.

(63) FIGS. **11** and **12** show that the fuel system frame **104** that includes the frame rail connection portions **106** (forward and rearward) can also include tank end support portion **258** that are secured to the first end **224** and the second end **232** of the fuel tank **102**. The tank support portions **258** extend across a diameter of and around forward and rearward ends of the fuel tank **102**. The tank support portions **258** on forward and rearward ends are connected by a structural member **256** to provide a concave frame structure into which the fuel tank **102** is received. The fuel system frame **104** can further include a collision load member **300** on a lateral side to absorb at least some of the load of an impact. The fuel system frame **104** can also include a lower frame assembly **254** for supporting the cover **105** and/or for enhancing the strength of the fuel system frame **104**. The lower frame assembly **254** can be disposed around and under a portion of the fuel tank **102** circumferentially between the positions of the tank support portions **258**. FIG. **11** shows that the fuel system frame **104** including the tank support portions **258**, the lower frame assembly **254** and the collision load member **300** can form an enclosure disposed about half of the circumference of the fuel tank **102** such that the fuel tank **102** can be placed within the frame from the other half circumference. In one assembly technique, the second tank support portion **268** is separated from the first tank support portion **260** of the tank support portion **258**. The first tank support portion **260** and the second tank support portion **268** can be separated on both ends of the frame assembly. The first boss **220** and the second boss **228** can be placed into the concave ends of the first tank support portion **260**. Thereafter, the second tank support portion **268** can be secured to the first tank support portion **260** to form the tank support portion **258** around the first boss **220** and the second boss **228**. Fasteners, such as bolts can then be advanced through apertures in the first tank support portion **260** and the second tank support portion **268** provide a rigid connection in the tank support portion **258**. In some embodiments, the lower frame assembly **254** is assembled to the tank support portion **258** on each end of the frame assembly after the second tank support portion **268** is secured to the first tank support portion **260**.

(64) FIG. **12** shows more detail of the structure of the tank support portion **258**. The tank support portion **258** includes a first tank support portion **260** and a second tank support portion **268**. The first tank support portion **260** can be disposed on a rearward side of the fuel system **100**. The second tank support portion **268** can be located on the forward side of the fuel system **100**. The first and second tank support portions **260**, **268** can be similar, e.g., mirror images of each other. The first tank support portion **260** will be discussed in detail. The description of the first tank support portion **260** and the second tank support portion **268** can be applied to each other.

(65) The first tank support portion **260** can include a first tank bracket **276** and a second tank bracket **280**. The first tank bracket **276** can include a first ribbed side **282** and a second ribbed side **284**. The first ribbed side **282** can have a plurality of, e.g., two, three, or four, or more than four ribs to enhance the strength of the first tank bracket **276**. The ribs can enhance the stiffness of the brackets.

(66) The second tank bracket **280** can include a first block **286** disposed at an end thereof configured to engage the fuel tank **102**. The first block **286** includes a first support surface **288** configured to engage the fuel tank **102**. The second tank bracket **280** also can include a second block **292**. The second block **292** includes a second support surface **294** configured to engage the fuel tank **102**. One or both of the first block **286** and the second block **292** can include a thicker portion of the first tank bracket **276** and the second tank bracket **280**, as shown in FIG. **11**. The blocks **286**, **292** can have a thickness sufficient to secure fasteners therein and therebetween. A space **296** defined between the first support surface **288** and the second support surface **294** can be configured to receive the first boss **220** of the fuel tank **102**. The first tank bracket **276** and the

second tank bracket **280** can be separate components to be joined as appropriate, e.g., by one or more bolts through the first block **286** and the second block **292**.

(67) FIGS. **11** and **12** show that the fuel system frame **104** can be configured to locate the mounting points or locations inward of the ends of the fuel system **100**. As discussed above, the first bracket **108** and the third bracket **140** are secured to the frame rail connection portions **106** away from the first end **224** and the second end **232** of the fuel tank **102**. For example, the first tank bracket **276** can be formed such that a first portion thereof extends to the first block **286** and a second end is coupled with or extends to the frame rail connection portion **106**. The first tank bracket **276** can comprise an L-shaped or curved construction with the first block **286** on one end and the frame rail connection portion **106** on an end opposite the first block **286**. The L-shape or curve can have an inner angle or portion that includes the first ribbed side **282**. The L-shape or curve can have an inner portion that follows the curvature of the fuel tank **102**. The inner angle or portion can be oriented toward the mounting space for the tank.

(68) FIGS. **11** and **12** show that this configuration allows the frame rail connection portion **106** to be located within the fuel system frame **104** at a longitudinal position of the fuel system **100** that is toward the central cylindrical portion **204** of the fuel tank **102**, e.g., at or over the cylindrical portion **204**. For example the second bracket **112** and/or the fourth bracket **152** can be aligned with, e.g., intersected by or centered on a plane containing the ends of the central cylindrical portion **204** (e.g., the liner portion) of the fuel tank **102**. In other words, a projection of the first end **224** and/or the second end **232** of the fuel tank **102** into the plane of the second portion **132** of the second bracket **112** would show the second bracket **112** to be between the projected ends. FIG. **2** shows that the second bracket **112** and the third bracket **140** can thus be spaced from the ends of the fuel system **100** with the cover **105** in place.

(69) The curved configuration of the first tank bracket **276** can allow the cover **105** to be shaped to create space for components on the frame rail **42**. As shown in FIG. **2** the shape of the first tank bracket **276** on the forward side of the fuel system **100** can enable the cover **105** to have an angled surface **103** creating space for the shackle **50** (or other low profile component mounted to the frame rail **42**). By moving the mount points on the frame rail connection portion **106** away from the forward and/or the rearward ends of the fuel system **100**, a more compact mounting arrangement or footprint for the fuel system **100** can be provided on the frame rail **42**. Also, the mount point locations on the frame rail connection portion **106** provide more options for supporting the fuel system **100** and other components on the frame rail **42**. For instance, the shackle **50** can be mounted to the frame rail **42** opposite the angled surface **103**, e.g., rearward of the projection **49** (see FIG. **2**).

(70) B. Extended Mounting Location Configurations

(71) While reducing the longitudinal extent of the footprint of the fuel system **100** between the second bracket **112** and the fourth bracket **152** can be beneficial, in some embodiments it is desirable to avoid locating mounting points between the ends of the fuel tank **102** or even between the forward and rearward faces of the cover **105**. FIGS. **11A** and **11B** show that the fuel system frame **104** can have another frame mounting configuration in which the first tank bracket **276** on the forward and/or the rearward ends of the fuel system **100** are re-oriented so that the first ribbed side **282** faces away from the space in which the fuel tank **102** is mounted.

(72) The first block **286** can be configured to mount to the second tank bracket **280** in at least two different orientations, e.g., as shown in FIG. **12** for a low profile frame rail mounting configuration or as in FIGS. **11A** and **11B** for a frame rail configuration avoiding a length A of the frame rail **42** opposite the fuel tank **102**. Fastening apertures on the first block **286** of the first tank bracket **276** and on the second block **292** of the second tank bracket **280** can align in a configuration where the internal angle of the first tank bracket **276** faces toward the fuel tank **102** or where the internal angle of the first tank bracket **276** faces away from the fuel tank **102**. In a configuration for reversing the first tank bracket **276**, mount features for supporting the lower frame assembly **254**

can be provided on both of the short sides that extend between the first ribbed side **282** and the second ribbed side **284**. The cover **105** can be modified to enclose the reversed first tank bracket **276** on one or both of the forward and rearward sides of the fuel system **100**. End portions of an overall enclosure of the fuel system **100** that includes the cover **105** can be modified to have an opening through which the first tank bracket **276** can extend to position the frame rail connection portion **106** at an exposed location for mounting to the frame rail **42**. For example, the cover **105** can be disposed circumferentially around the fuel tank **102**. Sheet or plate members can be coupled with the ends of the circumferential cover **105**. The sheet or plate members can cover then ends, e.g., the end shown in FIG. **13** to form an overall enclosure. When so extended the forward frame rail connection portion **106** is located forward of the second end **232** of the fuel tank **102** and the rearward frame rail connection portion **106** is located rearward of the first end **224** of the fuel tank **102**.

(73) The configuration of the fuel system frame **104** illustrated in FIGS. **11A** and **11B** provides an arrangement whereby a length of the frame rail **42** between the ends of the fuel system **100** and even between the first end **224** and the second end **232** of the fuel tank **102** is free of connection locations for the fuel system **100**. The fuel system **100** is disposed at or over this portion of the frame rail **42**. The frame rail **42** at the connection free length can be used for mounting other components between the fuel system **100** and the rail or on an in-board side of the frame rail **42** (opposite the side **44**). For example, internal trusses of the chassis of which the frame rail **42** is a part can be located along this length without concern for interference between such trusses and the brackets coupled to the forward and rearward frame rail connection portions **106**.

(74) III. Access Step Support Configurations

(75) In some cases it is desired to include a larger fuel tank **102**, e.g., a tank of more than 24 inches in diameter, e.g., 25 inches, 26 inches, or 27 inches in diameter. This desire for larger size of the fuel tank **102** conflicts with a limitation on the lateral extent of the fuel system **100** when applied to the vehicle **40**. With reference to FIG. **13**, it is desired to maximize dimension B (tank diameter) while not increasing dimension C (distance from frame rail mount location to outward most extent of the fuel system **100**). Another constraint is that if more than one step is provided, an upper step should be in-board of an upper step boundary **302**. The upper step boundary **302** provide enough clearance for a user to comfortably lift his or her foot from a lower step **324** to a step member **312** of a step support assembly **308**. In some embodiments, the fuel system **100** includes enhanced access step configurations that enable the fuel system **100** to be meet these conflicting requirements.

(76) FIGS. **2** and **13** shows that a step support assembly **308** can have an external portion disposed outside the cover **105** to be accessible to the user for accessing the cab of the vehicle **40**. The external portion of the step support assembly **308** can include a step member **312** that is formed to provide a low profile on a lateral side of the fuel system **100** and in some embodiments a second (e.g., a lower) step portion **324**. The step member **312** can be supported directly on an outside surface **320** the fuel tank **102** by a compressible member **316**, discussed further below. The direct support on the outside surface **320** of the fuel tank **102** helps to maintain a compact configuration in the width direction, e.g., to maintain as small a dimension C as possible (see FIG. **13**).

(77) Unlike many conventional steps, in some embodiments the step member **312** is not rectangular in cross-section. FIG. **13A** shows one profile of the step member **312** in which an upper surface **328** that is exposed and onto which the user can step. The upper surface **328** of the step member **312** can be generally horizontal to provide a flat surface for stepping. A first lateral edge **332** of the step member **312** can be disposed adjacent to the outside surface **320** of the fuel tank **102**. In one embodiment, the step member **312** can be formed from a sheet. An edge of the sheet can be folded to form the first lateral edge **332**. In particular, the edge of the sheet can be folded at least partially under the upper surface **328**. The folded under portion can be formed to generally follow the curvature of the tank. In one example, the folded under portion can have an internal angle of less

than 90 degrees or an external angle of greater than 270 degrees. The folded over portion can provide a length over which apertures for coupling to the compressible member **316** can be provided, as discussed further below. The step member **312** can have a second lateral edge **336** disposed opposite the first lateral edge **332**. The second lateral edge **336** can comprise the lateral most extent of the step member **312**. The second lateral edge **336** can be located inward of the upper step boundary **302**.

(78) The step member **312** can include a lower surface **340** that extends from a lower portion of the second lateral edge **336** toward the tank when assembled to the tank, as shown in FIG. **13A**. The lower surface **340** can be disposed at an angle to the second lateral edge **336**, e.g., at an internal angle that is greater than 90 degrees, or at an external angle that is greater than 180 degrees. As shown in FIG. **13A**, the lower surface **340** preferably extends at a non-horizontal angle toward the outside surface **320** of the fuel tank **102**. The internal angle between the lower surface **340** and the second lateral edge **336** is preferably about 120 degrees.

(79) As discussed above, the step member **312** can be formed from a sheet. In one embodiment, an edge of the sheet can be folded over to define the lower extent of the lower surface **340**. The folded over portion can include one or a plurality of apertures to facilitate securing the compressible member **316** and a portion of the cover **105** between the step member **312** and the fuel tank **102**. In one embodiment a folded over portion of the step member **312** extends away from the lower surface **340**. The folded over portion can extend generally in a direction along the fuel tank **102**. The folded over portion can be curved to follow the curvature of the fuel tank **102** or can be generally straight but disposed along a direction generally tangential to the outside surface **320** of the fuel tank **102**. The folded over portion can extend at an internal angle to the lower surface **340** of about 90 degrees. Other angles are possible. In some embodiment, the folded over ends of the step member **312** are sufficiently flexible to allow the ends to conform to the tank shape when the step member **312** is attached to the fuel system **100**.

(80) The non-rectangular shape of the step member **312** as formed and when applied to the outside surface **320** of the fuel tank **102** enable the fuel system **100** to maximize the length of the upper surface **328** while remaining in-board of the upper step boundary **302**. FIGS. **13** and **13A** illustrate one example of a non-rectangular step configuration that help maximize the size of the tank (dimension B—see FIG. **13**) while not overly extending the width dimension of the fuel system **100** (dimension C—see FIG. **13**).

(81) The step support assembly **308** also enhances compactness and step size by providing a more direct connection between the step member **312** and the fuel tank **102**. In one embodiment, folded over end portions of the step member **312** are mounted to the outside surface **320** through intervening compressible members **316**. The compressible member **316** are placed on an upper half of the fuel tank **102** such that a stepping force is opposed by the outside surface **320** of the fuel tank **102**. In particular, the load will be applied through the compressible members to the outside surface **320**. In one embodiment, the step member **312** is supported by one or a plurality of compressible member **316**, e.g., by two, four, six or eight compressible member **316**. The compressible member **316** can be formed of a polymeric material, such as one or more of a neoprene, a rubber material, nitrile rubber, natural rubber, and EPDM. The compressible member **316** can be generally circular in shape and can have a diameter of about one-half inch to two inches. In other specific examples, the compressible member **316** has a diameter of about one inch, about two inches, about three inches, or about four inches. FIG. **13** shows that the cover **105** can be disposed between the step member **312** and an outside surface of the compressible member **316**.

(82) The lower step **324** can be coupled with the collision load member **300**. In one embodiment, the cover **105** is disposed between the lower step **324** and the collision load member **300**. The lower step **324** can be intersected by the upper step boundary **302** such that a lower foot can rest on the lower step **324** and an upper foot of a user can be lifted up and over the second lateral edge **336** of the step support assembly **308** and come comfortably to rest across the upper step boundary **302**.

and onto the upper surface **328** of the step member **312**.

(83) In some embodiments the first bracket **108** can have a number of connection points to more permanently connect the first bracket **108** to the second bracket **112**. For example, a plurality of, e.g., two, apertures can be formed through upper segments of the first support wing **168** and the second support wing **172** to secure the first bracket **108** to the second bracket **112**.

(84) IV. Access Door Configurations

(85) In some cases it is desired to include an access door **400** on one or more ends of the fuel system **100**. The access door **400** can be configured to move between a closed position and one or more open positions. Moving the access door **400** to an open position can uncover an opening **420** on an end of the fuel system **100** and provide access to certain components within the fuel system **100**. In some cases, the opening **420** is vertically elongated to provided access along a majority, e.g., along 75 percent, of the height of the side of the cover **105**. The access door **400** can have a shape matching that of the opening **420**.

(86) In some embodiments, the access door **400** can be coupled to the cover **105** of the fuel system **100**. For example, as shown in FIGS. **14**, **14A**, and **14B**, the access door **400** can be rotatably coupled to an end portion of the cover **105** by a pivot **402**. In some embodiments, the pivot **402** is configured to provide some amount of resistance to rotation of the access door **400** (e.g., friction), thereby preventing the access door **400** from freely rotating about the pivot **402** under its own weight and allowing the access door **400** to remain fixed in any position. The resistance provided by the pivot **402** can be easily overcome by applying additional force to the access door **400**, such as by manually moving the access door **400** about the pivot **402**.

(87) In some embodiments, the axis of rotation of the access door **400** extends through the pivot **402**, perpendicular to the outer surface of the end portion of the cover **105**, allowing the access door **400** to be rotated between open and closed positions (e.g., rotated clockwise and/or counterclockwise) while keeping an inner surface of the access door **400** parallel to the outer surface of the end portion of the cover **105**. This configuration can advantageously allow the access door **400** to be opened and closed even when space is limited between components of the fuel system **100** (e.g., when an outwardly swinging door would not be usable). For example, in some embodiments, the access door **400** can be opened and closed when the fuel system **100** is side-mounted and there is 4 inches or less clearance between the fuel system **100** and another component mounted to the frame rail **42** adjacent to the system **100**. The access door **400** can be shaped such that the access door **400** does not interfere with portions of the fuel system frame **104** and brackets.

(88) In some embodiments, the access door **400** can be secured in a closed position using a fastener **410**. In some embodiments, the fastener **410** can comprise a bolt, rod, knob, lever, and/or button. For example, in some embodiments, the fastener **410** comprises a knob threaded on a rod. In some embodiments, the fastener **410** is fixed in a particular location on the fuel system **100** (e.g., does not change position when the access door **400** changes position). For example, the fastener **410** can be coupled to the outer surface of the cover **105**.

(89) In some embodiments, a portion of the access door **400**, such as a flange **404**, can be configured to be positioned between a lower surface of the fastener **410** and the outer surface of the cover **105** when the access door **400** is in the closed position. In some embodiments, the fastener **410** is configured to be tightened against a portion of the access door **400** to prevent rotation of the access door **400** about the pivot **402**. The flange **404** can comprise a hook portion that can be disposed around a shaft of the fastener **410** to at least temporarily fix the closed position of the access door **400**. For example, the fastener **410** can be configured to be rotated or otherwise adjusted to secure (e.g., compress) a portion of the access door **400**, such as the flange **404**, tightly against the outer surface of the cover **105**, thereby securing the access door **400** in position relative to the rest of the fuel system **100**. In some embodiments, rotation of the fastener **410** in a first direction (e.g., clockwise) tightens the fastener **410** against a portion of the access door **400** and

rotation of the fastener **410** in a second direction (e.g., counterclockwise), opposite the first direction, releases the access door **400** from its closed position, allowing the access door **400** to be rotated about the pivot **402**.

(90) In some embodiments, it is desired to include a holder **412** configured to hold the access door **400** in one or more positions, such as in an open position. For example, as shown in FIG. **14**, the holder **412** can be coupled to the cover **105** and configured to prevent the access door **400** from rotating beyond a particular location when the access door **400** is in an open position (e.g., by abutting an edge of the access door **400**). This can prevent the access door **400** from blocking or covering a portion of the opening **420** when the access door **400** is in an open position. The holder **412** can be a protrusion, rod, screw, and/or clamp.

(91) In some embodiments, the access door **400** can provide access, through the opening **420**, to three or more components within the fuel system **100**. This configuration can make it easier, faster, and/or more cost-efficient to service components of the fuel system **100**. For example, in some embodiments, as illustrated in FIGS. **14** and **14A**, the access door **400** can provide access to a filter **422**, a cylinder valve assembly **424**, a bleed valve **426**, a defuel assembly **430** comprising a defuel valve **432** and a defuel nozzle **434**, an electronic control unit (ECU) **428**, and/or fuel assembly plumbing.

(92) In some embodiments, the access door **400** can provide access, through the opening **420**, to components of the fuel system **100** that are mounted on more than one modular plate. The opening **420** can provide access to components of more than one subassembly. For example, the access door **400** can provide access to components of the fuel system **100** that are mounted on a first plate **440A**, a second plate **440B**, and/or a third plate **440C**. In some embodiments, the defuel assembly **430** and the filter **422** are mounted on different plates **440A**, **440C**. In some embodiments, the defuel assembly **430** and the ECU **428** are mounted on the same plate **440A**. In some embodiments, the filter **422** and the cylinder valve assembly **424** are mounted on the same plate **440C**.

(93) Although these inventions have been disclosed in the context of certain preferred embodiments and examples, it will be understood by those skilled in the art that the present inventions extend beyond the specifically disclosed embodiments to other alternative embodiments and/or uses of the inventions and obvious modifications and equivalents thereof. In addition, while several variations of the inventions have been shown and described in detail, other modifications, which are within the scope of these inventions, will be readily apparent to those of skill in the art based upon this disclosure. It is also contemplated that various combination or sub-combinations of the specific features and aspects of the embodiments may be made and still fall within the scope of the inventions. It should be understood that various features and aspects of the disclosed embodiments can be combined with or substituted for one another in order to form varying modes of the disclosed inventions. Thus, it is intended that the scope of at least some of the present inventions herein disclosed should not be limited by the particular disclosed embodiments described above.

Claims

1. A system for coupling a tank having a length, a first neck located on a first lengthwise end and a second neck located on a second lengthwise end, to a vehicle, the system comprising: a support assembly configured to connect the tank to a side portion of a frame rail of a vehicle, the support assembly comprising: a first tank end portion spanning an entire width of the tank and configured to receive a first neck of the tank, the first tank end portion having a first bracket and a second bracket, the first bracket configured to be coupled to a frame rail of the vehicle; and a second tank end portion spanning the entire width of the tank and configured to receive a second neck of the tank, the second tank end portion having a first bracket and a second bracket, the first bracket configured to be coupled to the frame rail of the vehicle.

2. The system of claim 1, wherein the first bracket of the first tank end portion and the first bracket

of the second tank end portion each have a frame rail connection portion and a neck portion, the frame rail connection portion being angled relative to the neck portion.

3. The system of claim 2, wherein the angled configuration of the frame rail connection portion and the neck portion form an interior side and an exterior side, a plurality of ribs configured to provide strength being disposed on the interior side.

4. The system of claim 3, wherein the plurality of ribs are located on the frame rail connection portion and the neck portion.

5. The system of claim 2, wherein the frame rail connection portion extends inwardly relative to the neck portion along a longitudinal direction.

6. The system of claim 2, wherein the frame rail connection portion extends outwardly relative to the neck portion along a longitudinal direction.

7. The system of claim 1, wherein the second bracket of the first tank end portion and the second bracket of the second tank end portion each have a first end and a second end, the first end coupled to the respective first bracket and the second end coupled to a collision load member configured to absorb at least a portion of a load of an impact.

8. The system of claim 7, wherein the collision load member spans a length of the tank.

9. The system of claim 7, further comprising a step coupled to the collision load member.

10. The system of claim 1, wherein the first bracket of the first tank end portion comprises a first block having a first support surface configured to be disposed about the first neck and the second bracket of the first tank end portion comprises a second block having a second support surface configured to be disposed opposite the first support surface and to enclose a space about the first neck.

11. The system of claim 1, wherein the first bracket of the second tank end portion comprises a first block having a first support surface configured to be disposed about the second neck and the second bracket of the second tank end portion comprises a second block having a second support surface configured to be disposed opposite the first support surface and to enclose a space about the second neck.

12. The system of claim 1, wherein the first tank end portion and the second tank end portion are angled relative to the frame rail of the vehicle.

13. The system of claim 12, wherein the first tank end portion and the second tank end portion are angled diagonally downward relative to the frame rail of the vehicle.

14. The system of claim 1, further comprising a cover configured to enclose a space around the tank.

15. The system of claim 14, further comprising a step coupled to the cover.

16. The system of claim 14, further comprising a door coupled to the cover and configured to rotatably move between a closed position and one or more open positions, the door covering an opening of the cover in the closed position.
